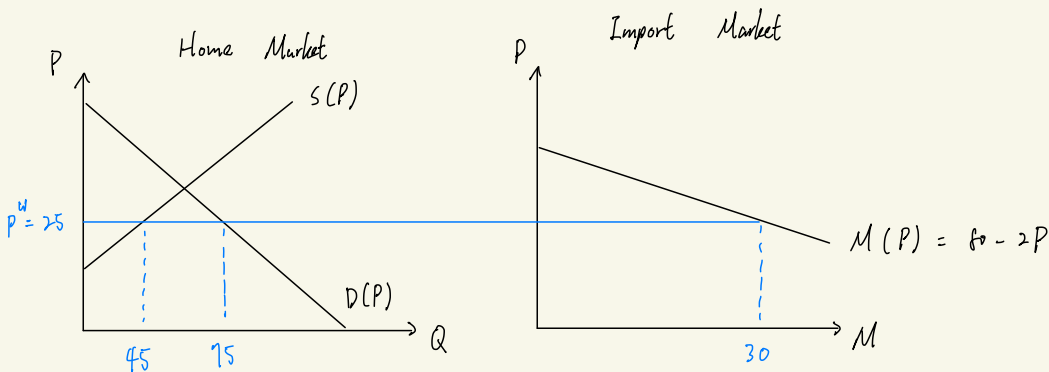


Practice 2 : The same tariff in a small and a large country

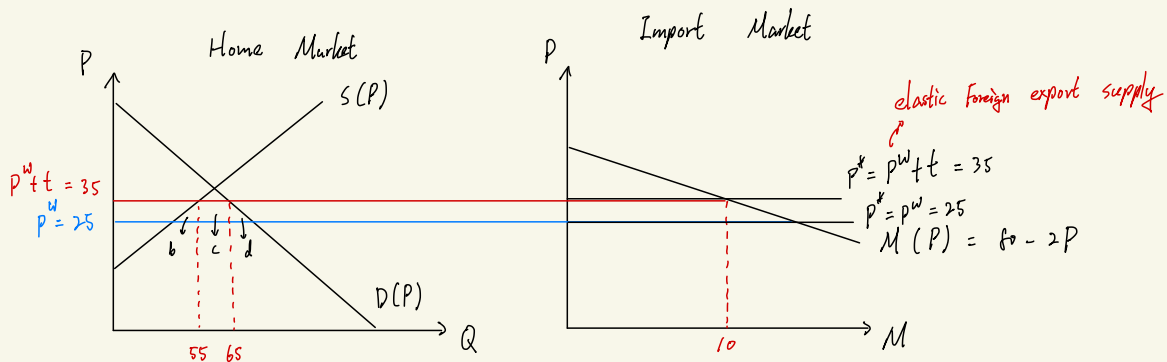
$$D(P) = 100 - P, \quad S(P) = 20 + P, \quad P^W = 25$$

$$M_i = D(P^W) - S(P^W), \quad t = 10$$



1. Home is a small country $\Rightarrow$ elastic foreign export supply

$$P^* = P^W = 25$$



Home price : $P^W + t = 35$

Domestic consumption : $D(P^W + t) = 65$

Domestic production : $S(P^W + t) = 55$

Imports after tariff : $M(P^W + t) = 10$

2.

$$\Delta CS = \text{triangle} - \text{triangle} = - \text{trapezoid} = - \frac{(65+75) \times 10}{2} = -700$$

$$\Delta PS = \text{triangle} - \text{triangle} = \text{trapezoid} = \frac{(55+45) \times 10}{2} = 500$$

$$TR = \text{rectangle} = 10 \times 10 = 100$$

$$\Delta WH = \Delta CS + \Delta PS + TR = -700 + 500 + 100 = -100$$

Transfer : Home consumers $\begin{cases} \rightarrow \text{Home suppliers } +500 \\ \rightarrow \text{Home government } +100 \end{cases}$
 -700

Distortion : ① Production Distortion

$$b : \text{triangle} = \frac{(55-45) \times 10}{2} = 50$$

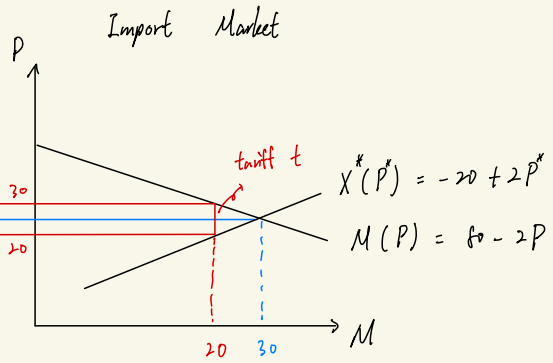
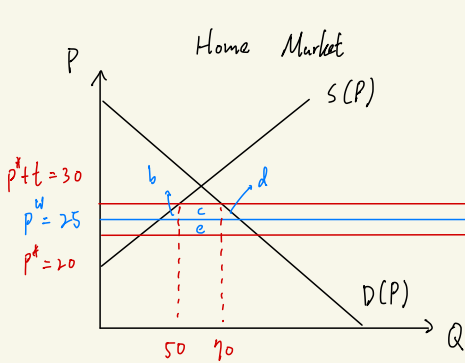
② Consumption distortion

$$d : \text{triangle} = \frac{(75-65) \times 10}{2} = 50$$

$$\Rightarrow b + d = 100$$

3. Home is a large country $\Rightarrow$ inelastic Foreign export supply

$$X^*(P^*) = -20 + 2P^*$$



Tariff creates the price wedge : $P - P^* = 10$

$\downarrow$ Home consumers pay
 $\downarrow$ Foreign suppliers receive

Market clearing condition : $M(P) = X^*(P^*)$

$$\Rightarrow 80 - 2(10 + P^*) = -20 + 2P^* \Rightarrow P^* = 20 \text{ \& } M = 20$$

$$4. \text{TR} = \begin{matrix} \boxed{\begin{matrix} c \\ e \end{matrix}} \begin{matrix} P \\ P^* \end{matrix} \\ \begin{matrix} 50 & 70 \end{matrix} \end{matrix} = 10 \times (70 - 50) = 200$$

TR here is higher than that in a small country due to "e".

$$b : \triangle = \frac{(50 - 45) \times (30 - 25)}{2} = 12.5$$

$$d : \triangle = \frac{(75 - 70) \times (30 - 25)}{2} = 12.5$$

$$\left. \begin{matrix} b \\ d \end{matrix} \right\} b + d = 25$$

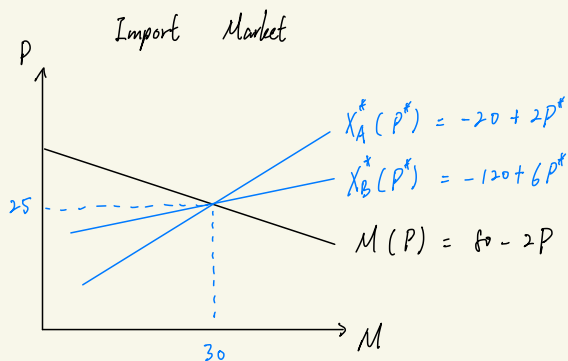
Total distortion becomes smaller

$$\Delta W_H = \underline{e} - (b + d) = 15$$

term of trade gain : Home pays Foreign less.

* The size of e depends on the elasticity of Foreign export supply

Practice 3 : Export supply elasticity



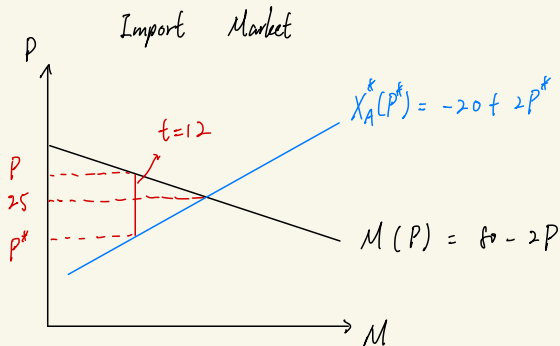
1. Elasticities of X_A^* & X_B^* at $(P, M) = (25, 30)$

$$E_A^* = \frac{dX_A^*/X_A^*}{dP^*/P^*} = \frac{dX_A^*}{dP^*} \times \frac{P^*}{X_A^*} = 2 \times \left(\frac{25}{30} \right) = \frac{5}{3}$$

$$E_B^* = \frac{dX_B^*/X_B^*}{dP^*/P^*} = \frac{dX_B^*}{dP^*} \times \frac{P^*}{X_B^*} = 6 \times \left(\frac{25}{30} \right) = 5$$

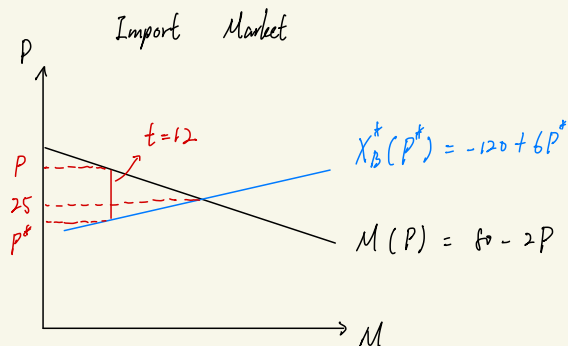
$X_B^*(P^*)$ is more elastic than $X_A^*(P^*)$ at $(25, 30)$

2. Import tariff $t = 12 \Rightarrow P = P^* + 12$



$$-20 + 2P^* = 80 - 2(P^* + 12)$$

$$\Rightarrow P^* = 19, P = 31, M = 18$$

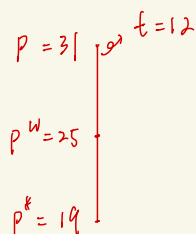


$$-120 + 6P^* = 80 - 2(P^* + 12)$$

$$\Rightarrow P^* = 22, P = 34, M = 12$$

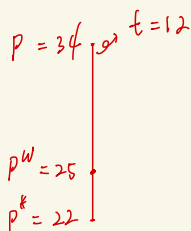
3. Pass-through : $\frac{P - P^W}{t}$ (The share of the tariff burdens in the importing country)

Case A :



$$\text{pass-through}_A = \frac{31 - 25}{12} = \frac{1}{2}$$

Case B :

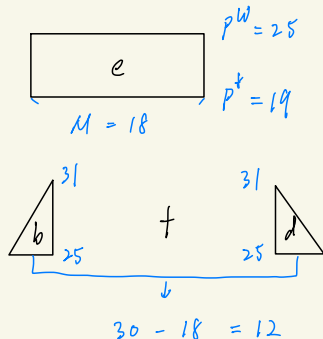


$$\text{pass-through}_B = \frac{34 - 25}{12} = \frac{3}{4}$$

* Takeaway : Tariff pass-through is higher when foreign export supply is more elastic, implying that a small country bears a larger share of the tariff burden.

→ Welfare changes : $\Delta W_H = e - (b + d)$
↑ term of trade ↓ distortion (or deadweight loss)
gain

Case A :



$$= (25 - 19) \times 18 = 108$$

$$= \frac{1}{2} (31 - 25) \times 12 = 36$$

$$\Delta W_{H,A} = 108 - 36 = \underline{72}$$

Case B :

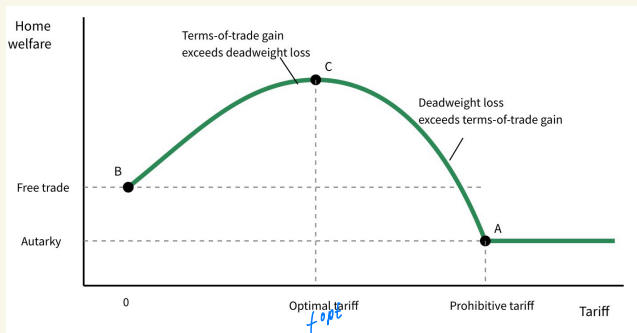
$$\begin{array}{c} \boxed{e} \\ \text{M} = 12 \end{array} \quad \begin{array}{l} p^W = 25 \\ p^t = 22 \end{array} = (25 - 22) \times 12 = 36$$

$$\begin{array}{c} \triangle b \quad \triangle d \\ \text{34} \quad \text{34} \\ \text{25} \quad \text{25} \end{array} + = \frac{1}{2} (34 - 25) \times 18 = 81$$

$30 - 12 = 18$

$$\Delta W_{H,B} = 36 - 81 = -45$$

* Takeaway : Compared with free trade, an import tariff may increase Home country's welfare when Foreign export supply is inelastic. This implies that an **optimal tariff** may exist for a large country.



$$W_H(t), \quad \left. \frac{dW_H(t)}{dt} \right|_{t=t^{opt}} = 0$$

$$\text{Optimal tariff } t^{opt} = \frac{P^*}{E^*}$$

$$\text{In the ad valorem term : } \tau = \frac{t}{P^*} \Rightarrow \tau^{opt} = \frac{1}{E^*}$$

$$\text{In a small country, } E^* \rightarrow \infty \Rightarrow \tau^{opt} = 0$$

$$\text{In a big country, } E^* < \infty \Rightarrow \tau^{opt} > 0$$

* Intuition : In a small country, perfectly elastic foreign export supply means that the importing country bears the entire tariff burden, so the optimal import tariff is zero.
tariff pass-through = 1